

10. Bootstrap Resampling (Statistical Significance)

Tests whether BLEU improvement is statistically significant.

$$\delta_b^* = \text{BLEU}(H_{\text{NER}}^{*b}, R^{*b}) - \text{BLEU}(H_{\text{base}}^{*b}, R^{*b}) , \quad b = 1, \dots, B$$

$$p = \frac{1}{B} \sum_{b=1}^B \mathbb{1}[\delta_b^* \leq 0], \quad \text{reject } H_0 \text{ if } p < \alpha \quad (B = 1000, \alpha = 0.05)$$